

DQ week one

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CSE 200: network and system administration

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1. What are the most critical responsibilities of a system administrator in a modern IT environment?

Ans: A system administrator or a system admin is an IT professional who is responsible for maintaining, configuring, and ensuring a reliable operation of computer systems that are particularly multi-users systems such as servers. The main goal of a system admin is to make sure the performance, resources and security of the systems they manage meet the demands of the users needs while staying under a budget (Azeus Convene,n.d.)

The responsibilities of a system administrator are:

1. Installing, configuring, and maintaining systems: Hardware and software components, operating systems such as Linux, Windows Server, or macOS, setup, configuration and management on servers and workstations.
2. System patching and updating: Ensuring systems are properly patched, updated and optimized for performance and security.
3. Planning and implementing upgrades: overseeing system upgrades to ensure the IT environment is up-to-date and effective.
4. Maintenance of networks and servers: Monitor and control the organisation's network of servers and workstations to ensure smooth running.
5. Supporting policy development for technology: in large organisations, the SysAdmin can also have some input into the technology policy, between technical operations and business objectives.

Reference

Azeus Convene. (n.d.). What is a system administrator? <https://www.azeusconvene.com/board-portal-glossary/system-administrator>

Kurose, J. F., & Ross, K. W. (2017). Computer networking: A top-down approach (7th ed.). Pearson.

2. How does the Linux philosophy of “everything is a file” influence system administration?

Ans: The "everything is a file" philosophy means Linux exposes devices, processes, sockets, and system information through a uniform file interface, and this shapes how sysadmins interact with a system in several important ways.

First, it makes administration tools composable. Since files can be read, written, and piped, commands like cat, grep, and redirection operators work on hardware devices in /dev, kernel parameters in /proc and /sys, just as they do on ordinary text files. An admin can check CPU info with cat /proc/cpuinfo or tune kernel settings by writing to /proc/sys, without needing specialized APIs for each subsystem.

Second, it simplifies permissions and security models. Because everything is a file, the same ownership and permission system (read, write, execute for user, group, others) applies uniformly across devices, sockets, and processes. This lets admins manage access control with a single consistent mechanism rather than juggling separate permission schemes per resource type.

Third, and directly relevant to networking, Linux represents network connections as sockets that behave like file descriptors. Kurose and Ross explain that a socket is the interface through which application processes send and receive data over the network, and in Unix systems this interface is implemented so that reading and writing to a socket uses the same system calls as reading and writing to a file (Kurose & Ross, 2017). This lets network programs reuse standard file I/O calls (read, write, close) instead of learning networking specific APIs, lowering the barrier for admins scripting network diagnostics or automating socket-based monitoring.

Tanenbaum and Bos, in Modern Operating Systems, similarly emphasize that treating devices and resources as files was a deliberate design choice in Unix to unify resource management and simplify the kernel's interface to user programs (Tanenbaum & Bos, 2014). This uniformity is what makes Linux scriptable and predictable at scale, a major reason it dominates server administration.

Reference

Kurose, J. F., & Ross, K. W. (2017). Computer networking: A top-down approach (7th ed.). Pearson.

Tanenbaum, A. S., & Bos, H. (2014). Modern operating systems (4th ed.). Pearson.

3. Compare the structure of the Linux file system with Windows. What are the advantages/disadvantages?

Ans: Linux and Windows organize storage in fundamentally different ways. Linux uses a single hierarchical tree rooted at / with drives and partitions mounted into this tree at points such as /home or /mnt, while Windows assigns each drive a separate letter (C, D) with its own independent root. Kurose and Ross explain that Unix like systems treat resources such as sockets uniformly as files, an approach that extends into how Linux exposes

storage, with everything accessible through one consistent namespace (Kurose & Ross, 2017). This means Linux handles devices, partitions, and network interfaces through the same file abstraction, whereas Windows keeps devices and file storage conceptually separate.

At the file system level, Linux commonly uses ext4, which relies on journaling, extent-based allocation, and delayed allocation to reduce fragmentation and speed up large file handling, while Windows primarily uses NTFS, which also journals but relies on a Master File Table and supports built in encryption and disk quotas. Sterniczuk's experimental study found that copying files under ext4 on Ubuntu was measurably faster than under NTFS on equivalent SSD hardware, attributing this to ext4's allocation strategy (Sterniczuk, 2022).

Advantages of Linux's structure include a unified namespace that simplifies scripting and permission management, and ext4's performance edge in write heavy workloads. Disadvantages include a steeper learning curve for users accustomed to drive letters, and fewer built in GUI tools for permission management. Windows advantages include tighter integration with EFS encryption and ACL based permissions, useful in enterprise environments, and a simpler mental model for casual users. Its disadvantages include weaker native scripting uniformity and comparatively slower file copy performance shown in benchmark testing.

References

- Kurose, J. F., & Ross, K. W. (2017). *Computer networking: A top-down approach* (7th ed.). Pearson.
- Sterniczuk, B. (2022). Comparison of EXT4 and NTFS filesystem performance. *Journal of Computer Sciences Institute*, 25, 297–300. <https://doi.org/10.35784/jcsi.3004>